

ABBA Method with Hairer's Symmetric Projection

Adaptation to GC2D with $H_\mu = 0$ and a constant simultaneous formulation

Complete replacement of BM4 by an A - B - B - A architecture

1 Guiding-center system and duplicated phase space

The physical state is

$$z = (X, Y) \in \mathbb{R}^2,$$

and, with the convention used here, it satisfies

$$\dot{X} = -\partial_Y \phi(t, X, Y), \quad \dot{Y} = \partial_X \phi(t, X, Y).$$

We define

$$J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad f(z, t) = J_0 \nabla_z \phi(z, t),$$

so that $\dot{z} = f(z, t)$. Its spatial Jacobian is

$$W(z, t) := D_z f(z, t) = J_0 \nabla_z^2 \phi(z, t).$$

Since the Hessian of ϕ is symmetric,

$$W(z, t)^T J_0 + J_0 W(z, t) = 0. \tag{1}$$

We duplicate the state:

$$\mathcal{Y} = (u, v) = (X_1, Y_1, X_2, Y_2) \in \mathbb{R}^4, \quad u, v \in \mathbb{R}^2.$$

The duplicated dynamics associated with the subflows used below is

$$\dot{u} = f(v, t), \quad \dot{v} = f(u, t).$$

If initially $u = v = z$, both equations coincide with the physical dynamics.

The physical state is identified with the diagonal manifold

$$\mathcal{M} = \{(u, v) \in \mathbb{R}^4 : u = v\}.$$

We write it as

$$g(\mathcal{Y}) = G\mathcal{Y} = u - v, \quad G = (I_2 \ -I_2), \quad GG^T = 2I_2.$$

For $\mu = (\mu_X, \mu_Y)^T$,

$$G^T \mu = \begin{pmatrix} \mu \\ -\mu \end{pmatrix} = (\mu_X, \mu_Y, -\mu_X, -\mu_Y)^T.$$

The constraints are linear. Therefore,

$$\nabla^2 g_1 = \nabla^2 g_2 = 0$$

and, for every μ ,

$$H_\mu(\mathcal{Y}) = \sum_{j=1}^2 \mu_j \nabla^2 g_j(\mathcal{Y}) = 0.$$

This does not mean that $\mu = 0$: H_μ is identically zero because $G = Dg$ is constant.

2 The explicit submaps A and B

For a substep length s and an evaluation time t , we define

$$A_{s,t}(u, v) = (u + sf(v, t), v), \quad (2)$$

$$B_{s,t}(u, v) = (u, v + sf(u, t)). \quad (3)$$

The map A advances only the first copy using the second one; B advances only the second copy using the first one. Both maps are explicit.

Their Jacobians are

$$DA_{s,t}(u, v) = \begin{pmatrix} I_2 & sW(v, t) \\ 0 & I_2 \end{pmatrix}, \quad DB_{s,t}(u, v) = \begin{pmatrix} I_2 & 0 \\ sW(u, t) & I_2 \end{pmatrix}. \quad (4)$$

In the duplicated phase space, we use the skew-symmetric matrix

$$\Omega_{\times} = \begin{pmatrix} 0 & J_0 \\ J_0 & 0 \end{pmatrix}.$$

Identity (1) directly implies

$$(DA_{s,t})^T \Omega_{\times} DA_{s,t} = \Omega_{\times}, \quad (DB_{s,t})^T \Omega_{\times} DB_{s,t} = \Omega_{\times}.$$

Thus, each submap is symplectic with respect to $\Omega_{\times}$, and any composition of these maps is also symplectic in the duplicated phase space.

Moreover, the variables used to evaluate the field remain fixed in each submap. Therefore,

$$\boxed{A_{s,t}^{-1} = A_{-s,t}, \quad B_{s,t}^{-1} = B_{-s,t}.} \quad (5)$$

3 The nonautonomous A – B – B – A step

Let

$$t_{n+1} = t_n + h, \quad s = \frac{h}{2}.$$

The base map that replaces BM4 is

$$\boxed{\Phi_{h,t_n}^{\text{ABBA}} = A_{s,t_{n+1}} \circ B_{s,t_{n+1}} \circ B_{s,t_n} \circ A_{s,t_n}.} \quad (6)$$

Compositions are read from right to left. Therefore, the effective order is

$$A_{s,t_n} \longrightarrow B_{s,t_n} \longrightarrow B_{s,t_{n+1}} \longrightarrow A_{s,t_{n+1}}.$$

Starting from $(u^{(0)}, v^{(0)})$, the four stages are

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \quad (A_1) \quad (7a)$$

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \quad (B_1) \quad (7b)$$

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \quad (B_2) \quad (7c)$$

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}). \quad (A_2) \quad (7d)$$

The output is

$$\Phi_{h,t_n}^{\text{ABBA}}(u^{(0)}, v^{(0)}) = (u^{(2)}, v^{(2)}).$$

The evaluations at the two time endpoints are essential in the nonautonomous problem. If every stage were evaluated at t_n , time symmetry would be lost. In the autonomous case, the two central stages can be combined, and one recovers

$$A_{h/2} \longrightarrow B_h \longrightarrow A_{h/2}.$$

Since A and B are symplectic with respect to $\Omega_\times$, the complete map is also symplectic:

$$\boxed{(D\Phi_{h,t_n}^{\text{ABBA}})^T \Omega_\times D\Phi_{h,t_n}^{\text{ABBA}} = \Omega_\times.} \quad (8)$$

4 Proof of the symmetry of ABBA

A time-dependent method is symmetric if

$$\Phi_{-h,t_{n+1}}^{\text{ABBA}} = \left(\Phi_{h,t_n}^{\text{ABBA}}\right)^{-1}. \quad (9)$$

Using (5) and reversing the order of composition in (6),

$$\left(\Phi_{h,t_n}^{\text{ABBA}}\right)^{-1} = A_{-s,t_n} \circ B_{-s,t_n} \circ B_{-s,t_{n+1}} \circ A_{-s,t_{n+1}}.$$

We now start at t_{n+1} with a step of size $-h$. Its final time is $t_{n+1} - h = t_n$, and definition (6) gives precisely

$$\Phi_{-h,t_{n+1}}^{\text{ABBA}} = A_{-s,t_n} \circ B_{-s,t_n} \circ B_{-s,t_{n+1}} \circ A_{-s,t_{n+1}}.$$

Therefore,

$$\boxed{\Phi_{-h,t_{n+1}}^{\text{ABBA}} = \left(\Phi_{h,t_n}^{\text{ABBA}}\right)^{-1}.}$$

The architecture is palindromic when both the sign of the step and the two endpoint times are reversed simultaneously.

5 Hairer's symmetric projection around ABBA

Suppose that

$$\mathcal{Y}_n = (z_n, z_n) \in \mathcal{M}$$

is known. For a multiplier $\mu \in \mathbb{R}^2$, we first correct the input:

$$\hat{\mathcal{Y}}_n(\mu) = \mathcal{Y}_n + G^T \mu = (z_n + \mu, z_n - \mu).$$

We write

$$u^{(0)} = z_n + \mu, \quad v^{(0)} = z_n - \mu,$$

and apply the four stages in (7). The auxiliary result is

$$\tilde{\mathcal{Y}}_{n+1}(\mu) = \Phi_{h,t_n}^{\text{ABBA}}(\mathcal{Y}_n + G^T \mu) = (u^{(2)}(\mu), v^{(2)}(\mu)).$$

The same correction is then applied at the output:

$$\mathcal{Y}_{n+1}(\mu) = \tilde{\mathcal{Y}}_{n+1}(\mu) + G^T \mu = (u^{(2)} + \mu, v^{(2)} - \mu). \quad (10)$$

Imposing $\mathcal{Y}_{n+1} \in \mathcal{M}$ is equivalent to

$$u^{(2)}(\mu) + \mu = v^{(2)}(\mu) - \mu.$$

Thus, the problem is reduced exactly to two equations:

$$\boxed{R(\mu) := u^{(2)}(\mu) - v^{(2)}(\mu) + 2\mu = 0.} \quad (11)$$

In matrix notation,

$$R(\mu) = G\Phi_{h,t_n}^{\text{ABBA}}(\mathcal{Y}_n + G^T\mu) + 2\mu. \quad (12)$$

Once the root μ_* has been found, the new physical state is

$$\boxed{z_{n+1} = u^{(2)}(\mu_*) + \mu_* = v^{(2)}(\mu_*) - \mu_*} \quad (13)$$

6 Exact Jacobian of the ABBA map

For a particular evaluation of (7), we define

$$\begin{aligned} W_1 &= W(v^{(0)}, t_n), & W_2 &= W(u^{(1)}, t_n), \\ W_3 &= W(u^{(1)}, t_{n+1}), & W_4 &= W(v^{(2)}, t_{n+1}), \end{aligned}$$

and abbreviate

$$S := W_2 + W_3.$$

The Jacobians of the four stages, evaluated at the points actually traversed, are

$$\begin{aligned} M_{A_1} &= \begin{pmatrix} I_2 & sW_1 \\ 0 & I_2 \end{pmatrix}, & M_{B_1} &= \begin{pmatrix} I_2 & 0 \\ sW_2 & I_2 \end{pmatrix}, \\ M_{B_2} &= \begin{pmatrix} I_2 & 0 \\ sW_3 & I_2 \end{pmatrix}, & M_{A_2} &= \begin{pmatrix} I_2 & sW_4 \\ 0 & I_2 \end{pmatrix}. \end{aligned}$$

By the chain rule,

$$\boxed{D\Phi_{h,t_n}^{\text{ABBA}} = M_{A_2}M_{B_2}M_{B_1}M_{A_1}.} \quad (14)$$

The two central matrices satisfy

$$M_{B_2}M_{B_1} = \begin{pmatrix} I_2 & 0 \\ sS & I_2 \end{pmatrix}.$$

Multiplying the four blocks gives the exact 4×4 Jacobian:

$$\boxed{D\Phi_{h,t_n}^{\text{ABBA}} = \begin{pmatrix} I_2 + s^2W_4S & s(W_1 + W_4) + s^3W_4SW_1 \\ sS & I_2 + s^2SW_1 \end{pmatrix}.} \quad (15)$$

The Jacobian of the map has not been replaced by the identity: (15) contains exactly all the dependencies generated by the four stages.

7 Exact Jacobian of the reduced residual

From

$$\frac{\partial(u^{(0)}, v^{(0)})}{\partial\mu} = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix} = G^T$$

and from (12), we obtain

$$\boxed{R'(\mu) = G D\Phi_{h,t_n}^{\text{ABBA}}(\mathcal{Y}_n + G^T\mu)G^T + 2I_2.} \quad (16)$$

If we write (15) as

$$D\Phi_{h,t_n}^{\text{ABBA}} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

then

$$G D\Phi_{h,t_n}^{\text{ABBA}} G^T = A - B - C + D.$$

Substituting the four blocks in (15) yields

$$\begin{aligned} R'(\mu) = & 4I_2 - s(W_1 + W_2 + W_3 + W_4) \\ & + s^2[W_4(W_2 + W_3) + (W_2 + W_3)W_1] \\ & - s^3W_4(W_2 + W_3)W_1. \end{aligned} \tag{17}$$

Since $s = h/2$, this can also be written as

$$\begin{aligned} R'(\mu) = & 4I_2 - \frac{h}{2}(W_1 + W_2 + W_3 + W_4) \\ & + \frac{h^2}{4}[W_4(W_2 + W_3) + (W_2 + W_3)W_1] \\ & - \frac{h^3}{8}W_4(W_2 + W_3)W_1. \end{aligned} \tag{18}$$

The matrices W_i need not commute: the order of the products in (17) and (18) must be preserved exactly.

8 Symmetry of the complete projected method

The equation defining the projected step can be written as

$$\mathcal{Y}_{n+1} - G^T \mu = \Phi_{h,t_n}^{\text{ABBA}}(\mathcal{Y}_n + G^T \mu), \quad \mathcal{Y}_n, \mathcal{Y}_{n+1} \in \mathcal{M}. \tag{19}$$

Applying the inverse of ABBA and using its symmetry,

$$\mathcal{Y}_n + G^T \mu = \Phi_{-h,t_{n+1}}^{\text{ABBA}}(\mathcal{Y}_{n+1} - G^T \mu).$$

We define the multiplier of the reverse step as

$$\mu_{\text{inv}} = -\mu.$$

Then

$$\mathcal{Y}_n - G^T \mu_{\text{inv}} = \Phi_{-h,t_{n+1}}^{\text{ABBA}}(\mathcal{Y}_{n+1} + G^T \mu_{\text{inv}}),$$

which is exactly equation (19) applied backward. Therefore, if the root is locally unique and is solved exactly,

$$\Psi_{-h,t_{n+1}} \circ \Psi_{h,t_n} = I. \tag{20}$$

Here Ψ denotes ABBA together with Hairer's projection. In an implementation, the iteration is stopped at a finite tolerance; reversibility is preserved up to that tolerance.

9 Simultaneous formulation and constant approximation

Alternatively, $\mathcal{Y}_{n+1}$ can be retained as an unknown. We define

$$d(\mathcal{Y}_{n+1}, \mu) := \mathcal{Y}_{n+1} - G^T \mu - \Phi_{h,t_n}^{\text{ABBA}}(\mathcal{Y}_n + G^T \mu).$$

We seek

$$d(\mathcal{Y}_{n+1}, \mu) = 0, \quad g(\mathcal{Y}_{n+1}) = 0.$$

Since G is constant and $H_\mu = 0$,

$$D_{\mathcal{Y}_{n+1}} d = I_4, \quad D_\mu d = - \left[I_4 + D\Phi_{h,t_n}^{\text{ABBA}}(\hat{\mathcal{Y}}_n) \right] G^T,$$

$$D_{\mathcal{Y}_{n+1}} g = G, \quad D_\mu g = 0.$$

The exact simultaneous Newton iteration solves

$$\boxed{\begin{pmatrix} I_4 & - \left[I_4 + D\Phi_{h,t_n}^{\text{ABBA}}(\hat{\mathcal{Y}}_n) \right] G^T \\ G & 0 \end{pmatrix} \begin{pmatrix} \Delta \mathcal{Y}_{n+1} \\ \Delta \mu \end{pmatrix} = - \begin{pmatrix} d(\mathcal{Y}_{n+1}, \mu) \\ g(\mathcal{Y}_{n+1}) \end{pmatrix}}. \quad (21)$$

Its Schur complement is

$$G \left[I_4 + D\Phi_{h,t_n}^{\text{ABBA}}(\hat{\mathcal{Y}}_n) \right] G^T = 2I_2 + GD\Phi_{h,t_n}^{\text{ABBA}}(\hat{\mathcal{Y}}_n)G^T = R'(\mu).$$

Therefore, the reduced and simultaneous formulations use exactly the same Jacobian.